

```

In[*]:= Modulo[{x_, y_}] := Sqrt[x^2 + y^2];
      raíz cuadrada

In[*]:= (*d es la abertura angular en cm. esc es el
      factor de escala en cm. rcm distancia tierra cm (pc)*)

In[*]:= rcm = 11;

In[*]:= esc = 3;

In[*]:= xycm[d_] := Tan[d * (1 / esc) * (1 / 3600) * (Pi / 180)] * rcm * (10^5);
      tangente      número pi

In[*]:= (*componentes x e y de las posiciones angulares*)

In[*]:= datad1cm =
      {{-4.4, 2.8}, {-5.2, -1.0}, {-1.45, -6.75}, {1.0, -7.5}, {7.8, -3.5}, {-3.7, 3.5},
      {-1.0, -6.7}, {6.8, 3.1}, {5.6, 3.9}, {2.4, 5.8}, {-4.4, 2.8}, {-5.5, 0.9}};

In[*]:= datad2cm = {{1.7, -1.2}, {2.5, 0.45}, {1.1, 3.4}, {-0.6, 3.9}, {-3.85, 1.8}, {1.7, -1.7},
      {0.4, 3.8}, {-3.6, -1.5}, {-3.1, -2.1}, {-1.3, -3}, {2.15, -1.4}, {2.25, -0.2}};

In[*]:= (*componentes x e y en distancia en 10-5 pc*)

In[*]:= datar1cm = xycm[datad1cm];

In[*]:= datar2cm = xycm[datad2cm];

In[*]:= (*Generar el dibujo de los puntos*)

In[*]:= f1 = ListPlot[datar1cm, Frame → True,
      representación de lista marco verdadero
      FrameLabel → {Style["xcm / 10-5 pc", Bold, 16], Style["ycm / 10-5 pc", 16, Bold]},
      etiqueta de marco estilo negrita estilo negrita
      BaseStyle → {FontFamily → "Calibri", FontSize → 14},
      estilo base familia de tipo de letra tamaño de tipo de letra
      PlotStyle → {Blend[{Red, Yellow, Black}], Thick}, PlotMarkers → {{ "x", 15}},
      estilo de repre... mezcla... rojo amarillo negro grueso marcadores de representación
      Epilog → {Black, PointSize@Large, Point[{0, 0}]},
      epílogo negro tamaño de ... grande punto
      PlotLegends → {"Observaciones m1"}, PlotRange → {{-10, 15}, {-15, 11}}];
      leyendas de representación rango de representación

In[*]:= f2 = ListPlot[datar2cm, PlotMarkers → {{ "x", 19}},
      representación de lista marcadores de representación
      PlotStyle → {Blend[{Blue, White}]}, PlotLegends → {"Observaciones m2"}];
      estilo de repre... mezcla... azul blanco leyendas de representación

In[*]:= (*Fit a elipses*)
      ajusta

In[*]:= g1 = ResourceFunction["EllipseFit"][datar1cm, {x, y}]
      función de recurso

Out[*]=
93.9772 + 3.40201 x - 0.687235 x2 - 1.95304 y - 0.0797177 x y - 0.722048 y2 == 0

```

```
In[*]:= ResourceFunction["EllipseProperties"][g1, {x, y}]
[función de recurso]
```

```
Out[*]:=
<|Equation → 93.9772 + 3.40201 x - 0.687235 x2 - 1.95304 y - 0.0797177 x y - 0.722048 y2 == 0,
Foci → {{6.06704, -3.78803}, {-0.943476, 0.800339}},
Vertices → {{12.8415, -8.22191}, {-7.71795, 5.23422}},
Covertices → {{8.88662, 8.1698}, {-3.76306, -11.1575}}, Eccentricity → 0.340987,
Center → {2.56178, -1.49385}, SemimajorAxisLength → 12.2858,
SemiminorAxisLength → 11.5494, AreaEnclosed → 445.772, SemiLatusRectum → 10.8573,
Directrices → {y == 60.3755 + 1.52789 x, y == -71.197 + 1.52789 x},
AxesOfSymmetry → {y == 0.182836 - 0.654498 x, y == -5.40797 + 1.52789 x},
Circumference → 74.8984, RotationAngle → -0.579531,
BaseConic → <|Equation → 0.00662516 x2 + 0.00749683 y2 == 1,
RotationAngle → -0.579531, DisplacementFromOrigin → {2.56178, -1.49385} |> |>
```

```
In[*]:= (*A1,B1,C1,D1,E1,F1*)
```

```
In[*]:= {A1, B1, C1, D1, E1, F1} = {-0.6872352716890875, -0.07971772430284296,
-0.7220476201597802, 3.4020093186182643, 1.953037084278905, 93.97715060564829};
```

```
In[*]:= g2 = ResourceFunction["EllipseFit"][datar2cm, {x, y}]
[función de recurso]
```

```
Out[*]:=
-23.2737 + 2.24049 x + 0.745741 x2 - 1.20118 y - 0.0454413 x y + 0.664684 y2 == 0
```

```
In[*]:= (*A2,B2,C2,D2,E2,F2*)
```

```
In[*]:= {A2, B2, C2, D2, E2, F2} = {0.7457412523130569, -0.04544129422509131,
0.6646841907084163, 2.240485498452707, 1.201175768269202, -23.27366707894891};
```

```
In[*]:= (*gráficas elipses*)
```

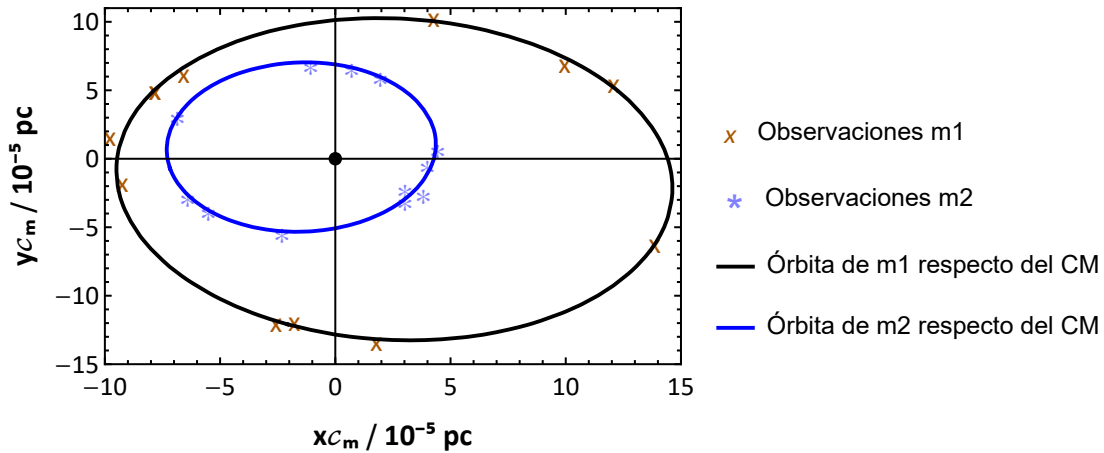
```
In[*]:= graphg1 = ContourPlot[g1, {x, -20, 20}, {y, -20, 20},
[representación de contornos]
ContourStyle → Black, PlotLegends → {"Órbita de m1 respecto del CM"}];
[estilo de contorno [negro] [leyendas de representación]
```

```
In[*]:= graphg2 = ContourPlot[g2, {x, -20, 20}, {y, -20, 20},
[representación de contornos]
ContourStyle → Blue, PlotLegends → {"Órbita de m2 respecto del CM"}];
[estilo de contorno [azul] [leyendas de representación]
```

```
In[*]:= Show[f1, f2, graphg1, graphg2]
```

```
[muestra
```

```
Out[*]=
```



```
In[*]:= (*B1'=0 → Angulo tal que este recto el eje de la elipse*)
```

```
In[*]:= FindRoot[(C1 - A1) * Sin[2 * x] + B1 * (1 - 2 * (Sin[x]) ^ 2) == 0, {x, 0}]
```

```
[encuentra raíz [seno [seno
```

```
Out[*]=
```

```
{x → -0.579531}
```

```
In[*]:= (*B2'=0 → Angulo tal que este recto el eje de la elipse*)
```

```
In[*]:= FindRoot[0 == (C2 - A2) * Sin[2 * x] + B2 * (1 - 2 * (Sin[x]) ^ 2), {x, 0}]
```

```
[encuentra raíz [seno [seno
```

```
Out[*]=
```

```
{x → -0.255476}
```

```
In[*]:= (*funciones que nos pasan al sistema del eje rotado pero recto*)
```

```
In[*]:= f[θ_, {A_, B_, Ca_, Da_, Ea_, F_}] :=  

  {A * (Cos[θ]) ^ 2 + B * Sin[θ] * Cos[θ] + Ca * (Sin[θ]) ^ 2,  

   [coseno [seno [coseno [seno  

   0, A * (Sin[θ]) ^ 2 - B * Sin[θ] * Cos[θ] + Ca * (Cos[θ]) ^ 2,  

   [seno [seno [coseno [coseno  

   Da * Cos[θ] + Ea * Sin[θ], -Da * Cos[θ] + Ea * Sin[θ], F}  

   [coseno [seno [coseno [seno
```

```
In[*]:= (*primera elipse rotada:*)
```

```
In[*]:= f[-0.5795307897171099, {A1, B1, C1, D1, E1, F1}]
```

```
Out[*]=
```

```
{-0.661148, 0, -0.748135, 1.77698, -3.91607, 93.9772}
```

```
In[*]:= (*segunda elipse rotada:*)
```

```
In[*]:= f[-0.25547579374506163, {A2, B2, C2, D2, E2, F2}]
```

```
Out[*]=
```

```
{0.751675, 0, 0.65875, 1.86422, -2.47131, -23.2737}
```

```
In[*]:= (*ecuaciones de las elipses rotadas*)
```

```
In[*]:= g1x = -0.6611477246129933 * x^2 - 0.7481351672358744 * y^2 +  

  1.7769837719257429 * x - 3.916072635808821 * y + 93.97715060564829 == 0;
```

```

In[*]:= g2x = 0.7516754981626701 * x^2 + 0.6587499448588032 * y^2 +
          1.8642223708156673 * x - 2.4713105094112344 * y - 23.27366707894891 == 0;

In[*]:= (*SEMIEJES MAJORES DE AMBAS ORBITAS*)

In[*]:= a1 = Sqrt[ ((1.7769837719257429 / (2 * Sqrt[-0.6611477246129933]))^2 +
  raíz cuadrada raíz cuadrada
  (-3.916072635808821 / (2 * Sqrt[-0.7481351672358744]))^2 -
  raíz cuadrada
  (93.97715060564829)) / (-0.6611477246129933) ]

Out[*]=
12.3166 + 0. i

In[*]:= a2 = Sqrt[ ((1.8642223708156673 / (2 * Sqrt[0.7516754981626701]))^2 +
  raíz cuadrada raíz cuadrada
  (-2.4713105094112344 / (2 * Sqrt[0.6587499448588032]))^2 -
  raíz cuadrada
  (-23.27366707894891)) / (0.7516754981626701) ]

Out[*]=
5.9652

In[*]:= (*-----*)
In[*]:= (*-----*)
In[*]:= (*-----*)

In[*]:= (*M1/M2*)

In[*]:= (*Normas de datar1cm*)

In[*]:= normr1cm = {Norm[{-7.821660722032338`, 4.977420459425207`}],
  norma
  Norm[{-9.243780853372746`, -1.7776501640698463`}],
  norma
  Norm[{-2.577592737903751`, -11.999138607936947`}],
  norma
  Norm[{1.7776501640698463`, -13.332376231165094`}],
  norma
  Norm[{13.865671280467103`, -6.221775574305396`}],
  norma
  Norm[{-6.577305607131093`, 6.221775574305396`}],
  norma
  Norm[{-1.7776501640698463`, -11.910256099723034`}],
  norma
  Norm[{12.088021116151015`, 5.510715508657827`}],
  norma
  Norm[{9.95484091905424`, 6.932835639958162`}],
  norma
  Norm[{4.266360393785309`, 10.310370951898069`}],
  norma
  Norm[{-7.821660722032338`, 4.977420459425207`}],
  norma
  Norm[{-9.77707590263311`, 1.599885147662597`}]};

In[*]:= (*Normas de datar2cm*)

```

```

In[*]:= normr2cm = {Norm[{3.022005278923711`, -2.133180196884632`}],
  Norm[{4.444125410194927`, 0.7999425738308754`}],
  Norm[{1.9554151804771887`, 6.0440105578930385`}],
  Norm[{-1.066590098441313`, 6.932835639958162`}],
  Norm[{-6.843953131751261`, 3.199770295331963`}],
  Norm[{3.022005278923711`, -3.022005278923711`}],
  Norm[{0.7110600656274186`, 6.755070623544449`}],
  Norm[{-6.399540590718076`, -2.6664752461076713`}],
  Norm[{-5.510715508657827`, -3.7330653445577595`}],
  Norm[{-2.3109452132921886`, -1100000 Tan[ $\frac{\pi}{648000}$ ]}],
  Norm[{3.821947852762222`, -2.4887102296998647`}],
  Norm[{3.999712869171299`, -0.3555300328136722`}]};

In[*]:= m1m2 = normr2cm / normr1cm
Out[*]=
{0.398988, 0.479706, 0.517602, 0.521503, 0.497119,
 0.47204, 0.56405, 0.521859, 0.548682, 0.520884, 0.491939, 0.405313}

In[*]:= (*MEDIA M1/M2*)
In[*]:= mediam1m2 = Total[m1m2] / 12
Out[*]=
0.494974

In[*]:= (*SEMIEJE RELATIVO*)
In[*]:= a = a1 + a2
Out[*]=
18.2818 + 0. i

In[*]:= (*-----*)
In[*]:= aenmetros = a * 10^(-5) * 3.0857 * 10^(16)
Out[*]=
5.64123 x 10^12 + 0. i

In[*]:= Paños = 79.419;
In[*]:= Psegundos = 79.419 * (365) * 24 * 3600
Out[*]=
2.50456 x 10^9

```

```
In[*]:= G = 6.6743 * 10^(-11);
```

```
In[*]:= m2 = (4 * Pi^2 * (aenmetros)^3) / (1.5 * G * (Psegundos)^2)
```

[Número pi](#)

```
Out[*]=
```

$1.12855 \times 10^{31} + 0. i$

```
In[*]:= m1 = mediam1m2 * m2
```

```
Out[*]=
```

$5.58603 \times 10^{30} + 0. i$

```
In[*]:= (*Masas respecto del Sol*)
```

```
In[*]:= m2M = m2 / (1.989 * 10^30)
```

```
Out[*]=
```

$5.67396 + 0. i$

```
In[*]:= m1M = m1 / (1.989 * 10^30)
```

```
Out[*]=
```

$2.80846 + 0. i$